VaaniShield

Product Requirements Document (PRD)

AI Project Submission — PRD

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1. Product Overview

VaaniShield is a research-grade detection system that flags likely voice-cloned segments in Indian code-switched (Hinglish-style) phone speech, aimed at financial-fraud call scenarios. It is positioned first as a rigorous academic contribution (a testable hypothesis about code-switch boundary artifacts, evaluated against existing baselines) and second as a proof-of-concept that could inform a future product for banks, NBFCs, or telecom fraud-prevention teams.

2. Problem & Opportunity (Summary)

Voice-cloning scams impersonating relatives or bank officials are rising sharply in India. Existing Indian-accent deepfake detectors and multimodal vishing-fusion systems do not account for intra-sentence code-switching, the linguistic register most common in the exact conversations fraud calls imitate. Full detail is in the companion Abstract & Problem Statement document.

3. Target Users & Use Cases

User	Use Case	Primary Need
Academic evaluators (faculty/reviewers)	Assess the project as a research submission	Clear novelty claim, rigorous evaluation against baselines
Bank / NBFC fraud analyst (future product user)	Review a flagged call with a risk score	Fast, explainable signal, low false-positive rate
Individual end user (stretch, future)	Live warning during a suspicious call	Real-time, low-latency, minimal setup

4. Goals & Success Metrics

- Demonstrate that the boundary-aware, speaker-nulled fusion model measurably outperforms Baseline A (generic WavLM/wav2vec2 classifier) and Baseline B (published Indian-accent Mel/LFCC/Wav2Vec model) on code-switched test data, reported via EER, precision/recall, and bootstrap confidence intervals.
- Show that the boundary-local stream and the speaker-nulled global stream each contribute independently via ablation (neither stream alone should match the fused result).
- Demonstrate reasonable generalization to a held-out (unseen during training) voice-cloning/TTS system.
- Produce a working demo (Streamlit or similar) that scores an uploaded/streamed call and shows an explainable, per-code-switch-point risk breakdown.

5. Scope

In scope:

- Hindi-English code-switched speech as the primary language pair; one additional regional-language pair (e.g. Kannada-English or Bengali-English) if time allows.

- Offline/batch detection on recorded or uploaded call audio for the evaluation and demo.
- A small, purpose-built dataset of bona-fide + cloned code-switched speech across multiple open voice-cloning systems.

Out of scope (for this submission):

- Production-grade real-time telecom/bank integration, on-device quantized deployment, and federated learning — these are established engineering (see SentinelEdge) and not this project's claimed contribution; only a lightweight demo is required.
- Coverage of every Indian language pair or every voice-cloning system on the market.
- Legal/regulatory rollout considerations for an actual telecom or banking deployment.

6. Functional Requirements

ID	Requirement	Priority
FR-1	System shall accept an audio file (or streamed audio) and its transcript as input.	Must
FR-2	System shall automatically identify code-switch points from the transcript and align them to audio timestamps.	Must
FR-3	System shall extract boundary-local prosodic/spectral discontinuity features at each code-switch point.	Must
FR-4	System shall extract a speaker-nulled global artifact residual from a self-supervised speech embedding.	Must
FR-5	System shall fuse both streams into a single fraud-risk score with a confidence interval.	Must
FR-6	System shall provide a human-readable explanation identifying which code-switch point(s) most influenced the score.	Must
FR-7	System shall report evaluation metrics (EER, precision/recall, CIs) broken down by language pair and noise condition.	Must
FR-8	Demo UI shall allow uploading a call recording and viewing the risk score and explanation.	Should
FR-9	System shall support at least two additional (held-out) voice-cloning systems for generalization testing.	Should

7. Non-Functional Requirements

- **Accuracy/rigor:** all reported results must include baselines and confidence intervals, not a single aggregate number.
- **Explainability:** every flagged call must surface which specific code-switch point(s) drove the score.
- **Reproducibility:** dataset generation scripts, model training code, and evaluation scripts must be version-controlled and re-runnable.
- **Privacy:** any real speaker recordings used for dataset construction must be from consenting participants; no real fraud-call recordings are used.
- **Usability (demo only):** the demo should return a result within a few seconds for a short call clip; hard real-time/edge latency is explicitly out of scope for this submission.

8. Assumptions & Constraints

- Open-source voice-cloning tools (Coqui XTTS, Bark, OpenVoice) are accessible and sufficient to generate a diverse enough spoof set within the project timeline.
- A small set of consenting speakers is available for bona-fide code-switched recordings; public corpora (e.g. IndicTTS, Common Voice Indian-language subsets) supplement this.
- Compute is limited to what's available for a one-month academic submission (no large-scale industrial training run) — model choices favor lightweight, fine-tunable components over training from scratch.

9. Milestones (High-Level)

Week	Milestone
1	Dataset construction: bona-fide + cloned code-switched recordings, code-switch annotation, noise augmentation.
2	Boundary-local feature pipeline built; Baseline A and Baseline B re-implemented for comparison.
3	Speaker-nulled global stream + fusion classifier built; ablation experiments run.
4	Full evaluation (EER/CIs/generalization), demo UI, and write-up finalized.

10. Risks

Risk	Mitigation
Too few consenting speakers to build a diverse enough bona-fide code-switched set.	Supplement with public Indian-language speech corpora; prioritize breadth of cloning systems over breadth of speakers.
Open-source voice-cloning tools may not naturally code-switch well, weakening the spoof set's realism.	Explicitly script code-switched prompts and manually verify a sample of generated clips before large-scale generation.
Boundary-local signal may turn out weaker than hypothesized on real data.	Ablation design surfaces this early (Week 3); framing the finding honestly either way is still a valid contribution.